



Self-healing bacteria killing gel

McMaster researchers have developed an unusual new gel made entirely from bacteria-killing viruses. <https://digit.in/aug19-39>



Neil Armstrong's wrongful death

A Cincinnati hospital paid Neil Armstrong's family \$6 million in a wrongful-death settlement. <https://digit.in/aug19-40>

have specific functions in the human cell. Simply tweaking the nucleobase (nitrogenous bases) code in RNA-based therapies, can adjust what impact the therapy has on patients. This makes RNA-based therapies more sought after since they transpire to be more targeted and versatile than other more common treatments such as immunotherapy, and DNA-based therapies.

USES OF RNA THERAPIES

Two main types of RNA treatment include antisense therapy which deteriorates and ultimately, destroys, malfunctioning or harmful proteins in a cell and mRNA which produces functional proteins. Multiple antisense therapies are already present that improve kidney transplant outcomes, treat haemophilia and lower cholesterol (LDL). One mechanism, known as RNAi (RNA interference), won the 2006 Nobel Prize for offering an array of new therapies. RNAi puts special kinds of short RNA to work that basically act as a set of specific instructions to the cell. A protein complex then utilises these instructions to destroy the mRNA that are malfunctioning or creating dysfunctional proteins. However, when clinical trials were conducted using this young innovation, the therapy caused severe side effects and the researchers needed to go back to the drawing board.

mRNA cancer vaccinations started being developed, and at the forefront is a company called BioNTech. They hope to create a universal cancer vaccine which fights all types of tumours. In Phase I of their trial, which was conducted in July 2017, 13 patients who had melanoma were treated using mRNA which contained 10 antigens that were unique to each person's tumour.

Another application of mRNA is in vaccinations for infectious diseases. It shows potential in many scenarios, right from the RSV virus which causes common cold to hepatitis and HIV. CureVac is one of the main players in this space. According to the CEO of CureVac, mRNA vaccines are more stable compared to ordinary ones since you can store them at room temperature.

This technology can also make vaccines as cheap as \$1 per shot. Researchers are now working on utilising RNA-encoding proteins from pathogens such as influenza, rabies or Zika.

On August 10, 2018, the FDA approved the first ever 'small interfering RNA' (siRNA) product. Alnylam Pharmaceuticals secured this approval and Orphan Drug Design for their siRNA product called Onpattro (patisiran). This drug treats a rare hereditary disease known as transthyretin-mediated amyloidosis in adult patients. The disease is caused due to the mutations in a protein which leads to neuropathic pain, loss of feeling in the limbs, and wheel-chair confinement. In Alnylam's clinical trial, during phase III, Onpattro managed to improve the manifestations of this disease and proved that the product was safe to use.



Onpattro for hereditary transthyretin-mediated amyloidosis

CHALLENGES IN IMPLEMENTATION

Despite the scientific promise and pharmaceutical companies' interest and investment, siRNA-based therapies are facing some hurdles, especially delivery. Delivery is proving to be difficult since siRNAs are quite large, negatively-charged molecules that naturally will not cross through a cell's outer membrane. Additionally, siRNAs are also excreted from the bloodstream quite rapidly when it is introduced into the human body. Also, siRNAs trigger innate immune receptors which initiate inflammation. This raises several questions about how exactly this siRNA therapy can be safely and selectively transported

to target tissues in the body and also avoid inflammation. Alnylam employed two different delivery approaches. The first is delivering siRNA molecules in a lipid nanoparticle since they naturally occur in the liver and can cross through the membrane of the cell. This method works well for protein targets that are found inside the liver. The approach was used by the company to deliver Onpattro to treat transthyretin-mediated amyloidosis since the transthyretin protein is largely produced in the liver.

However, late-stage siRNA therapeutic candidates were given the treatment by using another delivery approach. In this approach, the siRNA is attached to another molecule (such as sugar, receptor-ligand, an antibody fragment) which guides the siRNA to the target cell. Since this approach is not limited to the liver, a wider range of diseases can be targeted.

BENEFITS OF RNA THERAPY

The short and largely-specific arrangement of nucleic acids makes RNAi much easier to manufacture when compared to protein-based therapies, like immunotherapies. Longer strands, like mRNA, can be slightly harder, but there's tons of investment in their manufacturing. This essentially means that RNA therapy will most probably be much cheaper, stable and accessible than other complicated, personalised gene treatments. According to a recent study, RNA therapies also showed promise (in mice) to treat a very rare and difficult-to-treat disease known as spinal muscular atrophy (SMA).

Better vaccinations can be created, by dosing patients with mRNA that cause cells to change the strands into proteins and then stimulate the immune system to produce a response that is specific to the exact strain of the pathogen. This method can eliminate the need to inject live, dead or weakened pathogens in the current vaccines. Cancer cure, including universal cancer cures, are being developed aggressively. These therapies have shown sufficient success in clinical trials to warrant additional investment and hours being put in to develop the therapy techniques further. **4**